UNIT

III

ENGINEERING PHYSICS • DIGITAL NOTES

Dielectric and Magnetic Materials

Polarisation • Electric vectors • Electronic, ionic and orientation polarisation • Lorentz field • Clausius–Mossotti • Dielectric loss • Dia, para, ferro, antiferro, ferri magnetism • Domains • Hysteresis

QUICK SUMMARY • What you will learn

- How an insulator stores energy in an electric field and why its dielectric constant depends on frequency
- How the atomic magnetic moment leads to five classes of magnetic materials
- Why ferromagnets form domains, show hysteresis and are used as soft or hard magnets

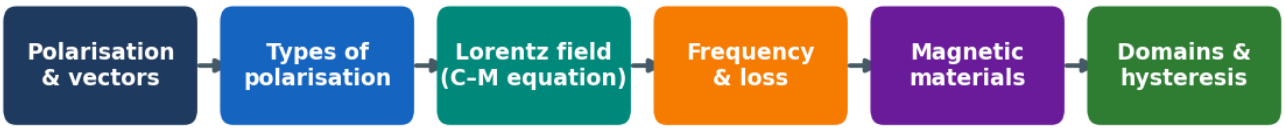


Fig. 1: Roadmap of Unit III

1 Dielectric Materials and Polarisation

DEFINITION • Dielectric

An insulating material with no free charges that can be **polarised** by an electric field. The field only shifts bound charges slightly, so no current flows. Examples: glass, mica, ceramics, polymers, water.

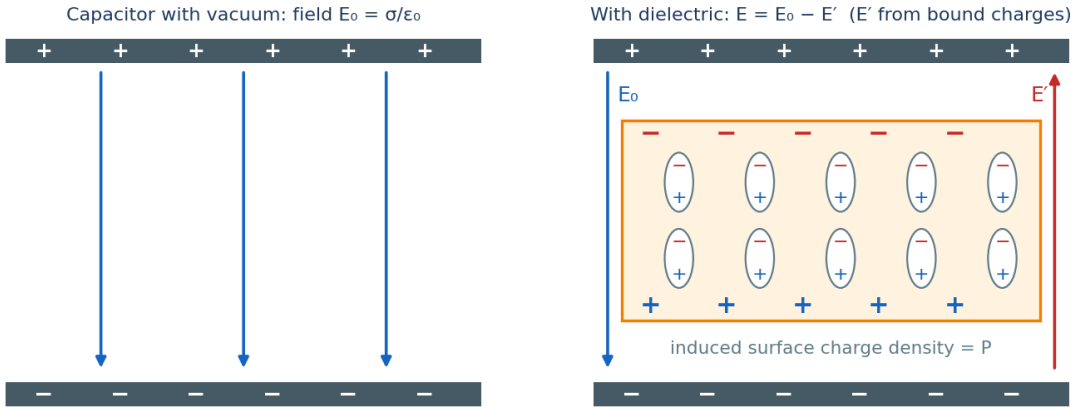


Fig. 2: A dielectric slab between capacitor plates: bound surface charges reduce the field inside

- **Dipole moment:** two equal and opposite charges ±q separated by a distance d form a dipole, μ = q d (C·m).
- **Dielectric polarisation (P):** dipole moment per unit volume, P = Nμ (C/m²). It equals the induced surface charge density.
- **Polar** molecules (H_2O, HCl) have permanent dipoles; **non-polar** molecules (H_2, N_2, CH_4) get induced dipoles only under a field.

Quantity	Symbol and definition	Unit
Polarisability	α = μ / E: induced dipole moment per unit field	F·m²
Polarisation	P = Nμ = NαE: dipole moment per unit volume	C/m²
Electric susceptibility	$\chi_e = P / (\varepsilon_0 E)$	dimensionless
Dielectric constant	$\varepsilon_r = \varepsilon/\varepsilon_0 = C_{medium}/C_{vacuum}$	dimensionless
Displacement vector	D = ε₀E + P: depends on free charge only (equals σ of plates)	C/m²

Relation between the electric vectors

Displacement vector	$D = \varepsilon_0 E + P = \varepsilon_0 \varepsilon_r E$
Polarisation	$P = \varepsilon_0 \chi_e E = \varepsilon_0 (\varepsilon_r - 1) E$
Susceptibility and ε_r	$\varepsilon_r = 1 + \chi_e$
Polarisation from atoms	$P = N\alpha E_i \quad (E_i = \text{local field})$

EXAMPLE • Numerical
A dielectric with $\varepsilon_r = 4$ is in a field $E = 10^5$ V/m. $\chi_e = 3$; $P = \varepsilon_0 \chi_e E = 8.85\times10^{-12} \times 3 \times 10^5 =$ **2.66×10^{-6} C/m²**; $D = \varepsilon_0 \varepsilon_r E =$ **3.54×10^{-6} C/m²**.

2 Types of Polarisation

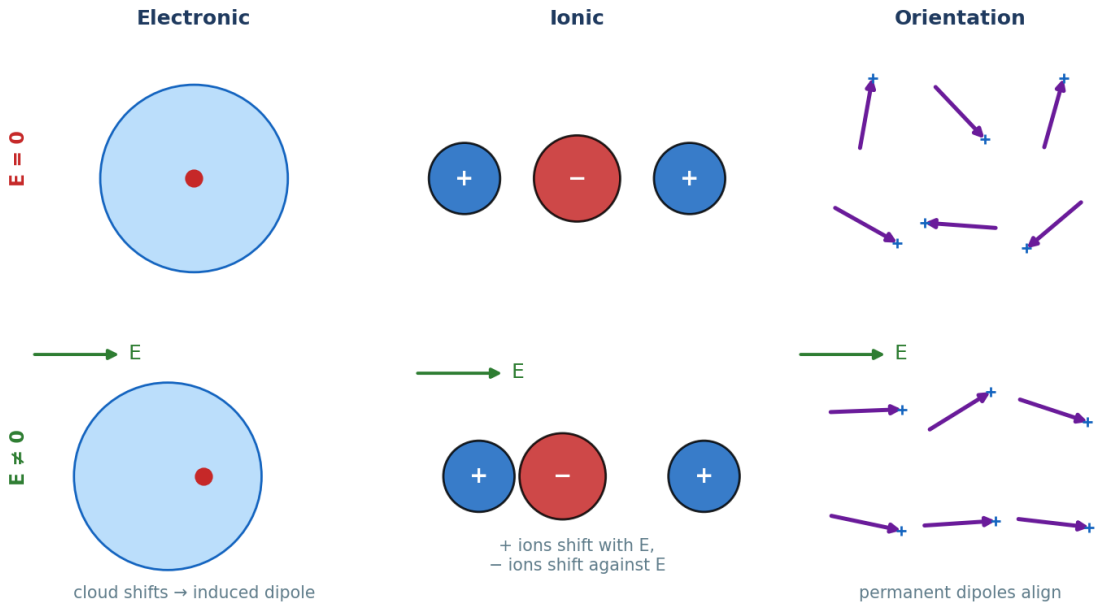


Fig. 3: Electronic, ionic and orientation polarisation without and with an applied field

Type	Cause	Present in	Time to set up	Temperature?
Electronic	Electron cloud shifts relative to nucleus	All materials	≈10^{-15} s (optical)	Independent
Ionic	+ and − ions displace in opposite directions	Ionic solids (NaCl, KCl)	≈10^{-12}–10^{-13} s (IR)	Almost independent
Orientation (dipolar)	Permanent dipoles line up with the field	Polar molecules (H_2O, HCl)	≈10^{-10} s or longer (micro/radio)	∝ 1/T
Space charge (interfacial)	Charges pile up at grain boundaries and interfaces	Impure / multiphase solids	Slow (audio and below)	Depends

Total polarisability	$\alpha = \alpha_e + \alpha_i + \alpha_o$

(a) Electronic polarisability (quantitative)

Model an atom as a nucleus +Ze at the centre of a uniform spherical electron cloud of radius R (total charge −Ze). Under field E the nucleus is displaced by x from the cloud centre. Only the charge inside radius x pulls back, so the restoring force is $F_r = Ze \times [Ze\, x / (4\pi\varepsilon_0 R^3)]$. Equilibrium: $F_r = ZeE$.

Displacement	$x = \frac{4\pi\varepsilon_0 R^3 E}{Ze}$
Induced dipole $\mu = Zex$	$\mu_e = 4\pi\varepsilon_0 R^3 E \Rightarrow \alpha_e = 4\pi\varepsilon_0 R^3$
Polarisation	$P_e = N\alpha_e E$

So α_e grows as R^3, and it is independent of temperature. Larger atoms are easier to polarise.

(b) Ionic polarisability (quantitative)

In an ionic pair (e.g. Na^+ Cl^-) the field shifts the positive ion by x_1 along E and the negative ion by x_2 against E. If β_1 and β_2 are the restoring-force constants, equilibrium gives $\beta_1 x_1 = eE$ and $\beta_2 x_2 = eE$.

Dipole moment $\mu = e(x_1 + x_2)$	$\mu_i = e^2 E\left(\frac{1}{\beta_1} + \frac{1}{\beta_2}\right)$
Ionic polarisability	$\alpha_i = e^2\left(\frac{1}{\beta_1} + \frac{1}{\beta_2}\right)$
With $\beta = m\omega_0^2$ (reduced mass m)	$\alpha_i = \frac{e^2}{\omega_0^2}\left(\frac{1}{m} + \frac{1}{M}\right)$

(c) Orientation polarisation (qualitative)

Polar molecules have permanent dipoles μ, normally randomised by thermal motion. A field gives each dipole a torque $\mu E \sin\theta$ that tries to align it, thermal agitation resists. For ordinary fields ($\mu E \ll kT$) the Langevin theory gives:

Orientation polarisability	$\alpha_o = \frac{\mu^2}{3kT} \qquad P_o = \frac{N\mu^2 E}{3kT}$

KEY IDEA • Temperature test

Only orientation polarisation falls as 1/T. So for polar dielectrics ε_r decreases when heated, while for non-polar dielectrics ε_r is nearly constant with temperature.

3 Internal (Lorentz) Field and Clausius–Mossotti Equation

An atom inside a solid feels the applied field **plus** the fields of all neighbouring dipoles. This **local (internal) field** E_i is larger than the macroscopic field E. Lorentz imagined a small sphere (cavity) around the atom: dipoles outside are treated as a continuum, those inside are summed individually.

Contributions to the field at the centre of the cavity:

- **E**: macroscopic field in the dielectric (applied field and the field of charges on the outer faces).
- **$E_2 = P/3\varepsilon_0$**: field due to polarisation charges on the cavity wall.
- **$E_3 = 0$**: field of the dipoles inside the cavity, which is zero for cubic symmetry or random arrangement.

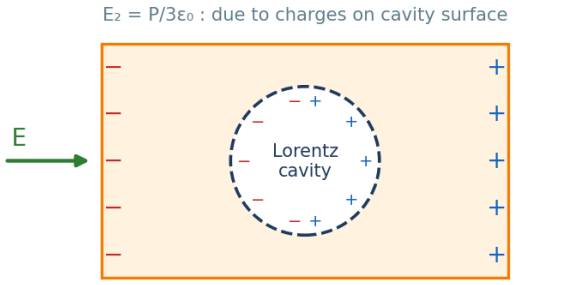


Fig. 4: Lorentz cavity in a polarised dielectric

Lorentz internal field	$E_i = E + \frac{P}{3\varepsilon_0}$

Substitute in $P = N\alpha E_i$ and use $P = \varepsilon_0(\varepsilon_r - 1)E$:

Clausius–Mossotti equation	$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = \frac{N\alpha}{3\varepsilon_0}$
Using $N = N_a\rho/M$	$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} \cdot \frac{M}{\rho} = \frac{N_A\alpha}{3\varepsilon_0}$

REMEMBER • Importance and limits

- It links a **macroscopic** property (ε_r) to a **microscopic** one (α), so α can be found from measured ε_r and density.
- Valid for elemental solids, non-polar gases and cubic crystals. Fails for strongly polar liquids such as water.

EXAMPLE • Numerical
A gas has N = 2.7 × 10^{25} m^{-3} and α = 2.2 × 10^{-40} F·m². Nα/3ε₀ = 2.24 × 10^{-4} = (ε_r−1)/(ε_r+2) → ε_r ≈ **1.00067**.

4 Complex Dielectric Constant, Frequency Dependence and Loss

Under an AC field the polarisation lags behind the field because dipoles need time to respond. The dielectric constant is therefore treated as a **complex** quantity.

Complex dielectric constant	$\varepsilon^* = \varepsilon' - j\varepsilon''$
Loss tangent	$\tan\delta = \frac{\varepsilon''}{\varepsilon'}$
Debye relaxation (τ = relaxation time)	$\varepsilon' = \varepsilon_\infty + \frac{\varepsilon_s - \varepsilon_\infty}{1+\omega^2\tau^2}, \quad \varepsilon'' = \frac{(\varepsilon_s - \varepsilon_\infty)\,\omega\tau}{1+\omega^2\tau^2}$
Power loss per unit volume	$P_{loss} = \omega\,\varepsilon_0\,\varepsilon'\tan\delta\,E_{rms}^2$

- ε′ (real part) = energy **stored**; ε″ (imaginary part) = energy **dissipated as heat** (dielectric loss).
- δ = loss angle, the angle by which the current in a real capacitor falls short of 90° ahead of voltage.

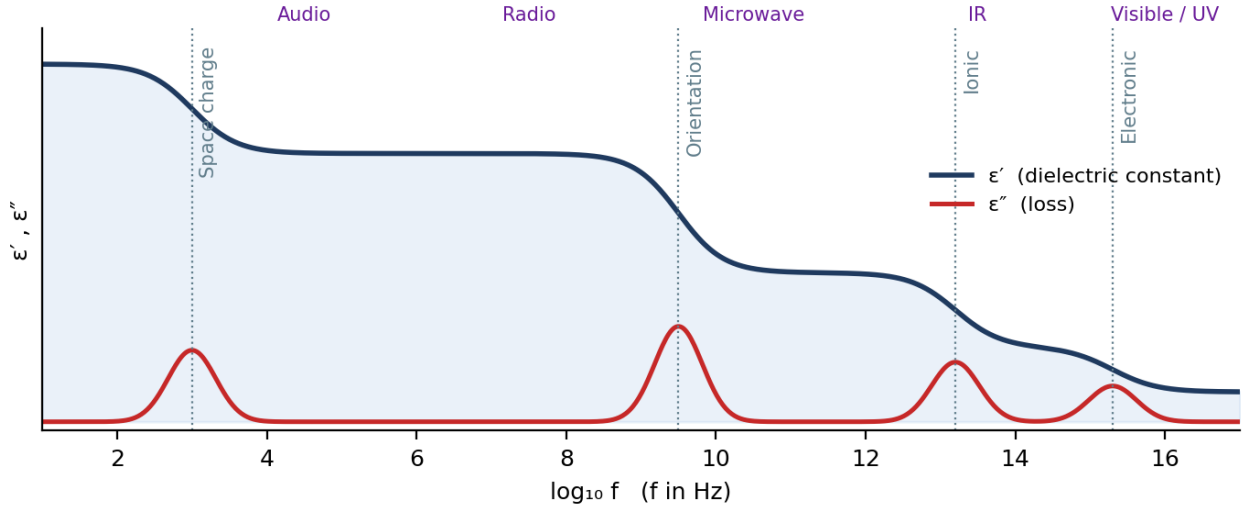


Fig. 5: Variation of ε′ and ε″ with frequency: each mechanism drops out when f exceeds its natural response rate, giving a step in ε′ and a peak in loss

- **Low frequency:** all four mechanisms follow the field, ε′ is largest.
- Above ~10^3 Hz space-charge polarisation can no longer follow; above ~10^9–10^{10} Hz (microwave) orientation polarisation drops out; above ~10^{13} Hz (IR) ionic drops out; only electronic remains up to ~10^{15} Hz (UV).
- At optical frequency $\varepsilon_r = n^2$ (refractive index squared) because only electronic polarisation survives.

APPLICATIONS • Dielectric loss matters in
- Capacitors: low tanδ is needed to avoid heating (mica, polystyrene).
- Microwave ovens: water's loss peak is used to heat food.
- Power cables and transformer oils: loss limits efficiency and causes breakdown.

5 Magnetic Materials: Basic Quantities

A magnetic dipole is a pair of poles ±p separated by l, or equivalently a current loop. Every magnetic property of matter arises from such moving charges.

Moment of a current loop	$m = IA$
Magnetisation (moment per volume)	$M = \frac{\sum m}{V}$
Flux density	$B = \mu_0(H + M) = \mu_0\mu_r H$
Susceptibility	$\chi = \frac{M}{H}, \qquad \mu_r = 1 + \chi$

Quantity	Meaning	Unit
Magnetic moment m	Strength of a magnetic dipole	A·m²
Magnetisation M	Magnetic moment per unit volume	A/m
Magnetic field H	Field due to external currents	A/m
Flux density B	Total field in the material	tesla (T)
Permeability μ	$\mu = B/H = \mu_0\mu_r$, with $\mu_0 = 4\pi \times 10^{-7}$ H/m	H/m

Atomic origin of magnetism

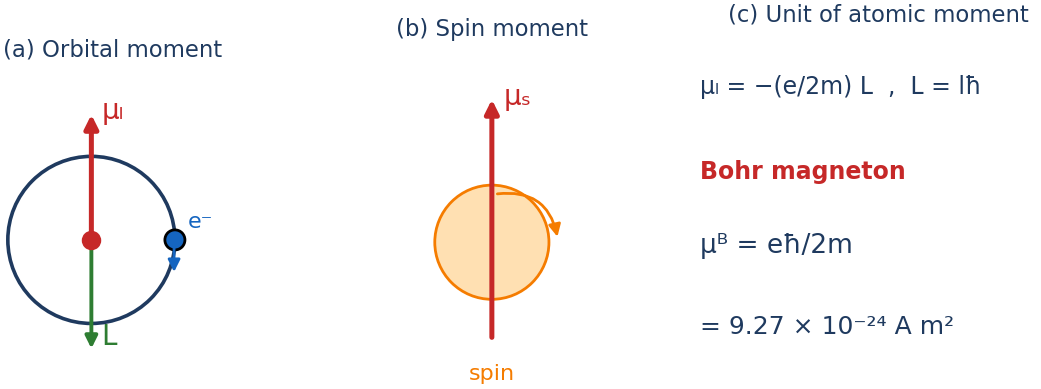


Fig. 6: Orbital motion and electron spin give atoms their magnetic moment

- **Orbital motion** of the electron around the nucleus acts like a tiny current loop and gives an orbital moment $\mu_l = -(e/2m)L$.
- **Electron spin** gives a spin moment of about one Bohr magneton; it is the main source in Fe, Co, Ni.
- **Nuclear spin** moments are about 10^3 times smaller and negligible.
- In filled shells the moments cancel in pairs (Pauli principle), so only atoms with **unpaired electrons** carry a net moment.

6 Classification of Magnetic Materials

Property	Dia	Para	Ferro	Antiferro	Ferri
Permanent atomic moment	None	Yes	Yes	Yes	Yes
Interaction	Induced only	Negligible	Strong, parallel (exchange)	Antiparallel, equal	Antiparallel, unequal
χ	Small, −ve	Small, +ve	Very large, +ve	Small, +ve	Large, +ve
μ_r	< 1	≈ 1	≫ 1	≈ 1	≫ 1
Temperature effect	None	$\chi = C/T$	Becomes para above T_c	Becomes para above T_N	Para above T_c
Example	Cu, Bi	Al, Pt	Fe, Co, Ni	MnO, Cr	Fe_3O_4

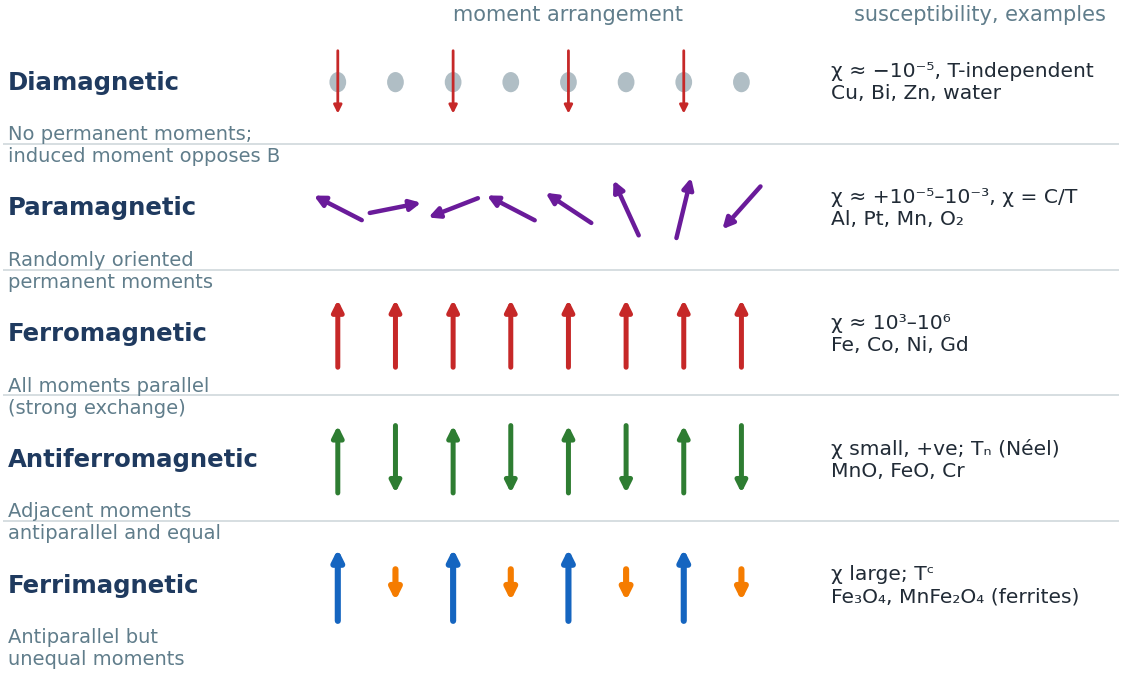


Fig. 7: Moment arrangement in dia-, para-, ferro-, antiferro- and ferrimagnetic materials

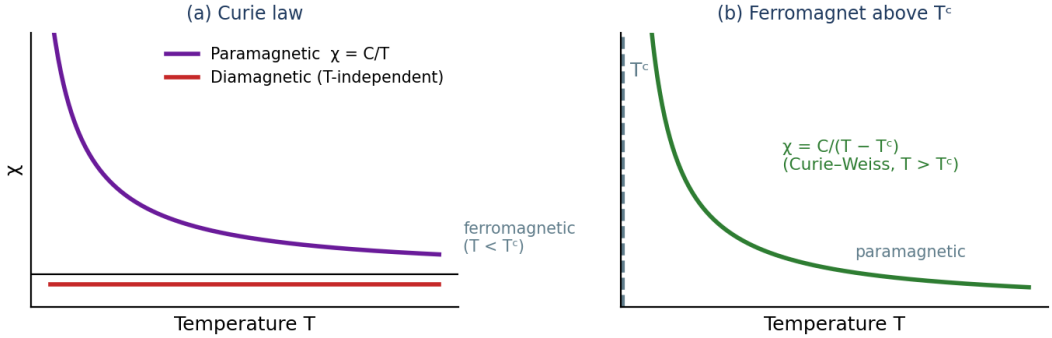


Fig. 8: Temperature dependence: Curie law and Curie–Weiss law

Curie law (paramagnet)	$\chi = \frac{C}{T}$
Curie–Weiss law (ferromagnet, $T > T^c$)	$\chi = \frac{C}{T - T_c}$

Curie temperatures: Fe 1043 K, Co 1388 K, Ni 627 K, Gd 292 K.

7 Domain Theory of Ferromagnetism and Hysteresis

Weiss proposed that a ferromagnet is divided into tiny regions, **domains** (about 10^{-6}–10^{-3} cm), each magnetised to saturation by exchange interaction. Without a field the domains point in different directions, so the net magnetisation is zero.

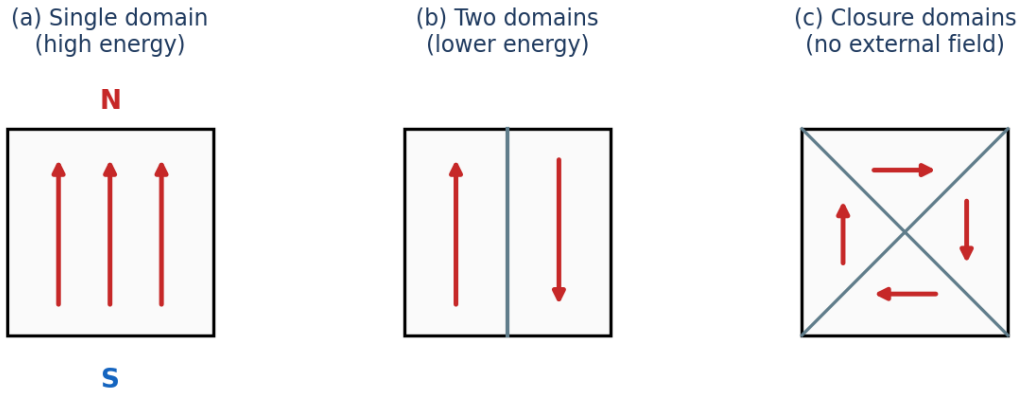
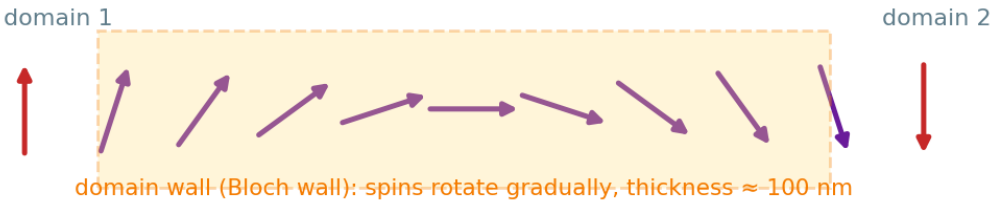


Fig. 9: Domains reduce magnetostatic energy: (a) single domain, (b) two, (c) closure domains with zero external field

DEFINITION • Domain wall (Bloch wall)
The transition region (~100 nm thick) between neighbouring domains in which the spin direction rotates gradually (by 180° or 90°). Gradual rotation lowers exchange energy, while wall thickness is limited by anisotropy energy.



- **Weak field:** domains already aligned with H grow by **wall displacement** (reversible).
- **Stronger field:** irreversible wall motion, then **domain rotation** aligns the rest; the material saturates.
- Domain structure is set by four energies: exchange, magnetostatic, anisotropy and magnetostrictive energy.

Hysteresis

When H is cycled, B lags behind H and traces a closed loop: **hysteresis**.

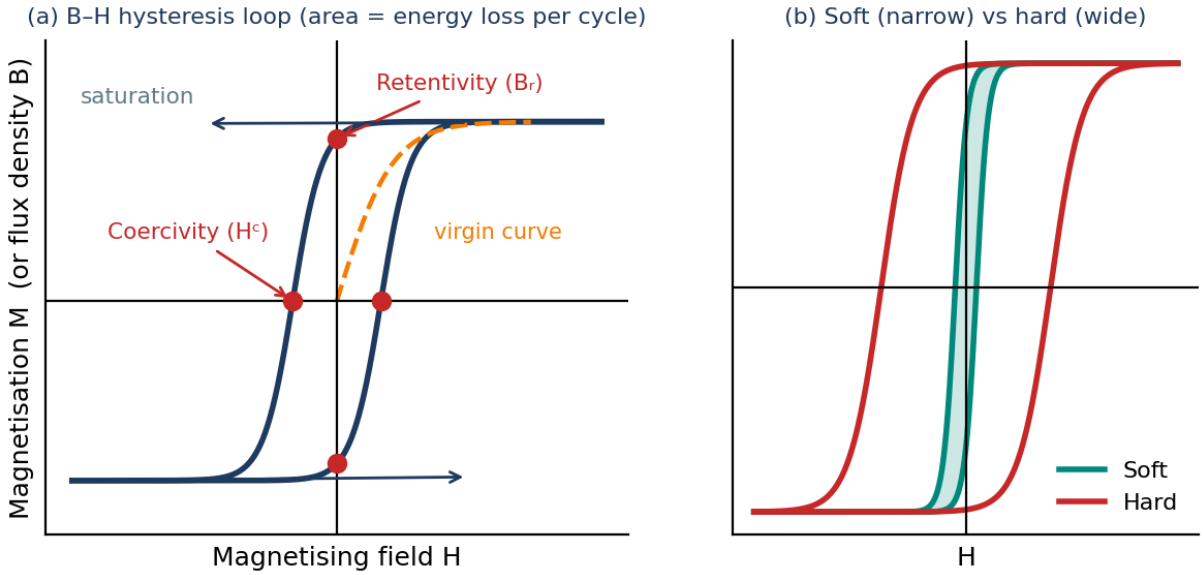


Fig. 10: Hysteresis loop and the comparison of soft and hard magnets

- **Retentivity (B_r):** flux density left when H returns to zero.
- **Coercivity (H_c):** reverse field needed to bring B to zero.
- **Loop area** = energy lost as heat per unit volume per cycle (hysteresis loss).

Property	Soft magnetic materials	Hard magnetic materials
Loop	Narrow, tall, small area	Wide, large area
Coercivity H_c, retentivity	Low H_c	High H_c and high B_r
Permeability	High	Low
Hysteresis loss	Small	Large
Magnetise / demagnetise	Easy	Hard
Materials	Silicon steel, permalloy, soft ferrites	Alnico, NdFeB, SmCo, hard ferrites
Uses	Transformer cores, motors, electromagnets, inductors	Permanent magnets, loudspeakers, magnetic storage

KEY IDEA • Exchange interaction (Weiss field)
Ferromagnetism needs a strong quantum **exchange interaction** between neighbouring spins that favours parallel alignment (equivalent to an internal "Weiss molecular field" of about 10^3 T). Above T_c thermal energy beats exchange and the order disappears.

APPLICATIONS • Applications of magnetic materials

- **Soft:** transformer and motor cores (silicon steel), inductors and high-frequency cores (soft ferrites), magnetic shielding (permalloy).
- **Hard:** permanent magnets in loudspeakers, motors, MRI, generators (NdFeB, SmCo, alnico).
- **Ferrites:** high resistivity → low eddy-current loss at microwave frequencies; memory cores, magnetic tapes and hard disks.

APPLICATIONS • Applications of dielectrics

- Capacitors (high ε_r ceramics like $BaTiO_3$ give large capacitance), insulation of cables, transformer oil.
- Piezoelectric and ferroelectric devices: sensors, ultrasound transducers, memory.
- Microwave resonators and substrates (low-loss ceramics).

8 Worked Problems

EXAMPLE • 1. Electronic polarisability

For an atom of radius R = 1.5 Å: $\alpha_e = 4\pi\varepsilon_0 R^3 = 1.113\times10^{-10} \times (1.5\times10^{-10})^3 = \mathbf{3.76 \times 10^{-40}\ F\cdot m^2}$.

EXAMPLE • 2. Loss tangent

A dielectric has $\varepsilon' = 5.0$ and $\varepsilon'' = 0.01$. $\tan\delta = 0.01/5.0 = \mathbf{2 \times 10^{-3}}$; at 1 MHz and $E_{rms} = 10^4$ V/m the loss is $P = \omega\varepsilon_0\varepsilon''E^2 = 2\pi\times10^6 \times 8.85\times10^{-12} \times 0.01 \times 10^8 \approx \mathbf{55.6\ W/m^3}$.

EXAMPLE • 3. Bohr magneton

$\mu_B = e\hbar/2m = (1.6\times10^{-19} \times 1.055\times10^{-34}) / (2 \times 9.11\times10^{-31}) = \mathbf{9.27 \times 10^{-24}\ A\cdot m^2}$.

EXAMPLE • 4. Curie–Weiss

Iron ($T_c = 1043$ K) has $\chi = 5 \times 10^{-3}$ at 1100 K. $C = \chi(T - T_c) = 5\times10^{-3} \times 57 = 0.285$ K; at 1200 K, $\chi = 0.285/157 = \mathbf{1.8 \times 10^{-3}}$.

Key terms at a glance

Term	Meaning
Bound charge	Charge held to its atom, can only shift slightly
Relaxation time τ	Time for polarisation to reach 1/e of its final value after a field change
Exchange energy	Quantum energy that makes neighbouring spins parallel (ferro) or antiparallel (antiferro)
Magnetostriction	Change of dimensions of a magnetic material when it is magnetised
Eddy current	Circulating current induced in a conductor by changing flux, causing heat loss

9 Formula Sheet and Exam Corner

Topic	Key formulae
Electric vectors	$D = \varepsilon_0 E + P = \varepsilon_0\varepsilon_r E$; $P = \varepsilon_0\chi_e E$; $\varepsilon_r = 1 + \chi_e$
Polarisabilities	$\alpha_e = 4\pi\varepsilon_0 R^3$; $\alpha_i = e^2(1/\beta_1 + 1/\beta_2)$; $\alpha_o = \mu^2/3kT$
Lorentz field	$E_i = E + P/3\varepsilon_0$
Clausius–Mossotti	$(\varepsilon_r - 1)/(\varepsilon_r + 2) = N\alpha/3\varepsilon_0$
Loss	$\varepsilon^* = \varepsilon' - j\varepsilon''$; $\tan\delta = \varepsilon''/\varepsilon'$
Magnetic	$B = \mu_0(H + M)$; $\chi = M/H$; $\mu_r = 1 + \chi$; $\mu_B = e\hbar/2m = 9.27\times10^{-24}$ A·m²
Laws	Curie: $\chi = C/T$; Curie–Weiss: $\chi = C/(T - T_c)$

EXAMPLE • Numerical

A paramagnetic material ($\chi = 2.3 \times 10^{-5}$) is in $H = 10^6$ A/m. $M = \chi H = \mathbf{23\ A/m}$; $B = \mu_0(H + M) = 4\pi\times10^{-7} \times (10^6 + 23) \approx \mathbf{1.2566\ T}$.

EXAM CORNER • Short questions (2–3 marks)

- Define dielectric polarisation, polarisability and susceptibility.
- Write the relation among D, E and P.
- What are the different types of polarisation?
- What is the Lorentz internal field? Write the Clausius–Mossotti equation.
- Define dielectric loss and loss tangent.
- Define Bohr magneton.
- Distinguish between soft and hard magnetic materials.
- What is a domain wall?

EXAM CORNER • Long questions (10 marks)

- Derive expressions for electronic and ionic polarisability.
- Explain orientation polarisation and its dependence on temperature.
- Derive the Clausius–Mossotti equation using the Lorentz internal field.
- Discuss the frequency dependence of polarisation and dielectric loss.
- Classify magnetic materials with the help of neat diagrams and examples.
- Explain the domain theory of ferromagnetism and the hysteresis loop.

EXAM CORNER • Quick MCQs

- 1. Which polarisation depends on temperature? (a) electronic (b) ionic (c) orientation.
- 2. Local field in a cubic crystal: (a) $E + P/3\varepsilon_0$ (b) $E - P/3\varepsilon_0$ (c) E.
- 3. Hard magnets have (a) low H_c (b) high H_c and B_r.
- 4. Antiferromagnets become paramagnetic above (a) Curie (b) Néel temperature.
- **Answers:** 1-c, 2-a, 3-b, 4-b.

QUICK SUMMARY

- Dielectric: $D = \varepsilon_0 E + P$; polarisation = electronic + ionic + orientation + space charge, each cutting off at its own frequency.
- Clausius–Mossotti connects ε_r to α using the Lorentz field $E + P/3\varepsilon_0$.
- Magnetism comes from electron orbit and spin; domains and hysteresis explain ferromagnets, and soft/hard materials suit different devices.